

SMART INDIA HACKATHON 2026



TITLE PAGE

PROBLEM STATEMENT METADATA:

Problem Statement ID — 26050

Problem Statement Title — High Altitude Performance Optimization and Robust Design of Anti-Drone System

Project Solution Title — HIGH-ALTITUDE RUGGEDIZED ANTI-DRONE DETECTION & PRECISION TRACKING SYSTEM WITH AI-BASED ENVIRONMENTAL COMPENSATION

Theme — Robotics and Drones / Defence & Security

PS Category — Software / Hardware Integration (Hybrid)

Team ID — SIH2026-PS26050-TEAM

Team Name — HAADS SIH 26050



IDEA TITLE

❖ PROBLEM (HIGH-ALTITUDE OPERATIONAL STRESS):

- Extreme cold (-30° C) affects motor pulse response, lubricant viscosity, and mechanical cable stiffness.
- Low atmospheric pressure (500 hPa) changes air density and dynamic aerodynamic behaviour.
- High wind gusts (60 km/h) cause severe aerodynamic drag (2.55 N) and target-pointing drift.
- Mountain glare and atmospheric haze reduce visual sensor reliability.
- Uncompensated environmental variation causes tracking instability and target acquisition loss.

PROPOSED SOLUTION (INTEGRATED HARDWARE + AI EDGE VISION):

- Real-time YOLO26n-based visual object detection (sub-40ms edge inference).
- SORT-based persistent target tracking maintaining track IDs with center locking (320, 240).
- BME280 environmental monitoring (Temperature + Atmospheric Pressure + Humidity).
- MPU6050 6-axis IMU for tilt, motion, and structural vibration monitoring.
- Anemometer wind-speed sensor for real-time aerodynamic drag compensation.
- Environment-aware compensation engine scaling servo pulse signals (Stiffness factor x1.5).
- ESP32 DevKit V1 telemetry communication through HiveMQ MQTT broker.
- 2-axis Pan-Tilt servo mechanism for physical target-centred tracking.
- Piezo buzzer local alert indication (GPIO 25 @ 2000Hz tone).

INNOVATION & INTEGRATED CHAIN:

Environment sensing + AI edge detection + persistent tracking + dynamic compensation + pan-tilt servo control in one unified system.

🚀 INTEGRATED FLOW: SENSORS (BME280/MPU6050/Wind) → AI DETECTION (YOLO26n) → TARGET TRACKING (SORT) → COMPENSATION ENGINE → PAN-TILT SERVO CONTROL → TARGET LOCK

- **1. REAL PHYSICAL HARDWARE COMPONENTS (PROTOTYPE BOM):**

- ESP32 DevKit V1: Main 32-bit dual-core microcontroller, sensor acquisition, PWM servo control, MQTT Wi-Fi telemetry.
- BME280 Sensor Module: Ingests ambient Temperature, Atmospheric Pressure, and Relative Humidity.
- MPU6050 IMU Module: Measures 3-axis acceleration, tilt/orientation angles, and structural vibration.
- Anemometer / Wind Speed Sensor: Real wind-speed measurement used for aerodynamic drag force compensation.
- SG90 / MG90S Servo Motors: 2 precision servo motors (One for PAN azimuth, One for TILT elevation).
- 2-Axis Pan-Tilt Camera Bracket: Mechanical gimbal mounting structure holding and moving the camera payload.
- USB Webcam: Real camera visual input for physical target detection and tracking.
- Piezo Buzzer: Local audible alert output indicating target lock and system events (GPIO 25).
- 10kΩ Potentiometers: Manual testing and threshold calibration inputs.
- 5V Regulated Power Supply & Power Bus: Provides clean DC power for servos and microcontroller peripherals.
- Breadboard & Jumper Wires: Circuit wiring and inter-module signal connections.

2. COMPUTING & AI EDGE PLATFORM (LAPTOP / SYSTEM):

- Laptop / Computer: Runs Python 3.12, OpenCV 4.x, YOLO26n Edge AI model, SORT Tracker, and Streamlit Dashboard.

3. COMMUNICATION ARCHITECTURE:

- ESP32 DevKit —[Wi-Fi / HiveMQ MQTT Broker]—> Python Hardware Interface —> Streamlit Dashboard

4. SIMULATION VALIDATION LINK (WOKWI):

- WOKWI SIMULATION: Virtual ESP32 platform used for hardware-in-the-loop validation before physical wiring.

5. END TO END SYSTEM DATA FLOWS:

FEASIBILITY AND VIABILITY



- **TECHNICAL FEASIBILITY:**

- ESP32 DevKit provides highly reliable, low-cost embedded control for sensor sampling and servo PWM generation.
- BME280, MPU6050, and anemometer supply accurate environmental telemetry for dynamic software compensation.
- 2-axis servo pan-tilt mechanism enables clear physical target tracking demonstration.
- Laptop CPU/GPU handles real-time YOLO26n Edge AI vision and SORT tracking during the prototype phase.
- Wokwi simulation enables safe virtual hardware testing before physical assembly.
- Modular system design supports seamless transition to standalone embedded edge hardware (e.g. Jetson Orin).

OPERATIONAL CHALLENGES:

- Extreme cold (-30° C) increasing cable resistance, lubricant viscosity, and motor torque demand.
- Low atmospheric pressure (500 hPa) altering air density and thermal dissipation efficiency.
- High wind disturbance (60 km/h) inducing aerodynamic drag force (2.55 N) and pointing drift.
- Mechanical vibration and chatter caused by high-altitude winds.
- Sensor drift and visual glare under intense high-altitude sunlight.
- Servo positioning backlash and potential MQTT communication network latency.

MITIGATION STRATEGIES:

- Temperature-aware pulse compensation multiplying servo drive gain (Stiffness factor $\times 1.5$).
- Wind-speed based aerodynamic drag calculation and automatic pan/tilt angular offset correction.
- IMU-based vibration monitoring and adaptive stabilization filtering.
- Sensor validation and noise rejection algorithms.
- SORT persistent object tracking preserving target Track IDs through momentary camera glare drops.
- Servo position feedback control and MQTT heartbeat liveness monitoring.
- Truthful 9-subsystem health monitor providing complete operator visibility.

IMPACT AND BENEFITS



- **TARGET APPLICATIONS & USER DOMAINS:**

- High-altitude security installations and remote defense outposts (Himalayan border sectors).
- Mountainous border surveillance and perimeter monitoring corridors.
- Critical national infrastructure protection (power plants, communication relays, airbases).
- Defense R&D laboratories and prototyping testbeds.
- Non-kinetic anti-drone detection, tracking, and authorized response research.

SYSTEM BENEFITS:

- Environment-aware tracking resilient against severe atmospheric and thermal stress.
- Improved pointing stability and reduced target acquisition loss under strong wind gusts.
- Real-time environmental monitoring (Temperature, Pressure, Wind Speed, Vibration).
- Real-time operator visibility into complete system health via 9-subsystem health matrix.
- Modular hardware architecture allowing straightforward component upgrades and maintenance.
- Low-cost simulation-first development model combining Wokwi ESP32 and open-source AI tools.
- Scalable edge-AI architecture designed for easy future transition to rugged field deployment.

SCALABILITY ROADMAP:

- CURRENT PROTOTYPE: Laptop + USB Webcam + Wokwi ESP32 Simulation & MQTT Telemetry Link.
- PHYSICAL PROTOTYPE: ESP32 + BME280 + MPU6050 + Anemometer + USB Webcam + 2-Axis Servos + Buzzer + 5V Supply.
- FUTURE FIELD DEPLOYMENT: Embedded Edge AI (Jetson Orin) + Ruggedized IP67 Enclosure + High-Altitude Field Operations.

RESEARCH AND REFERENCES



- **GENUINE TECHNICAL & RESEARCH REFERENCES:**

- [1] YOLO / Ultralytics Documentation: Real-time object detection models and edge vision deployment benchmarks.
- [2] SORT Tracking Algorithm (Bewley et al.): Simple Online and Realtime Tracking for persistent target centroid locking.
- [3] ESP32 Microcontroller Documentation (Espressif Systems): ESP32 dual-core architecture, PWM, and Wi-Fi stack.
- [4] Bosch BME280 Sensor Datasheet: Environmental sensing principles for barometric pressure, temperature & humidity.
- [5] MPU6050 IMU Datasheet (InvenSense): 6-axis motion tracking, accelerometer, and gyroscope processing.
- [6] Anemometer Measurement Principles: Principles of wind-speed measurement and aerodynamic drag calculation.
- [7] Wokwi Simulator Documentation: Virtual microcontroller hardware simulation platform for ESP32 and IoT modules.
- [8] MQTT Protocol Specification (OASIS Standard): Lightweight publish/subscribe messaging transport for IoT telemetry.
- [9] MIL-STD-810H Military Standard: Environmental engineering considerations and laboratory tests for high altitude & extreme cold.